

Muhammad Talha Zaheer

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PROFESSIONAL SUMMARY

Cybersecurity professional and technical educator passionate about mentoring the next generation of security practitioners. Experienced in penetration testing, defensive security, digital forensics, and hands-on lab instruction—from building isolated Kali environments to teaching OWASP Top 10 exploitation and remediation. Designed and delivered multi-module video curricula for Skill Bridge Pakistan and Hunarmand Punjab, and foster local talent as Hack The Box Meetup Ambassador and Cybersecurity Mentor at UMT. Grounded in a BS in Computer Science plus diplomas in Cyber Security and Digital Forensics, with a teaching approach built on *think like an attacker, defend like a professional*.

TECHNICAL SKILLS

Security Domains	Penetration Testing, Incident Response, Reconnaissance & Footprinting, Network Traffic Analysis, Network Security, Web Application Security, OWASP Top 10, OSINT, Digital Forensics, Vulnerability Assessment, Threat Modeling
Offensive & Defensive Tools	Kali Linux, Parrot OS, Metasploit Framework, Nmap, Burp Suite, SQLmap, WPScan, Wireshark, Hashcat, Ghidra, Autopsy, Maltego, Nessus, Network Miner, DVWA, pfSense, WAF, VirtualBox, Netcat
Languages	C++, Python, Bash, SQL, JavaScript
Instruction & Leadership	Curriculum Design, Workshop Delivery, CTF Organization, Team Leadership, Mentorship

PROFESSIONAL EXPERIENCE

Hack The Box <i>Meetup Host & Ambassador</i>	<i>Apr 2025 – Present</i> <i>Lahore, Pakistan</i>
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- Hosted on-campus and online cybersecurity meetups at NIIT, UCP, FAST Lahore, UMT Lahore, Superior University, ShellShift, and virtual sessions, teaching security fundamentals through workshops and hands-on labs.
- Mentor aspiring professionals on career paths, certifications, and practical skill development; provide HTB resources and curated learning support.
- Expand community outreach as Lahore Ambassador (Jan 2026–Present), fostering collaborative peer learning environments.

Spacebar Training <i>Cyber Security Trainer</i>	<i>May 2026 – Present</i> <i>Lahore, Pakistan</i>
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- Compiled a 12-module cybersecurity curriculum outline spanning beginner to expert levels, designed for UK students.
- Wrote voice-over scripts for cyber security lectures and recorded lecture videos with practical demonstrations for the UK-targeted course.

Skill Bridge Pakistan <i>Cyber Security Course Instructor</i>	<i>Dec 2025 – May 2026</i> <i>Lahore, Pakistan</i>
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- Designed and recorded a 24-lecture hands-on curriculum teaching attack-to-defend methodology from foundational labs to industry reporting.
- Taught isolated lab setup (VirtualBox/Kali), Wireshark traffic analysis, Nmap reconnaissance, Burp Suite web testing (DVWA), and EternalBlue (MS17-010) exploitation.

- Covered OSINT, Google Dorking, WAF remediation, digital forensics basics, and vulnerability reports with executive summaries and remediation steps.

Hunarmand Punjab

Cyber Security Course Instructor

Jul 2025 – Oct 2025

Punjab, Pakistan

- Designed two full courses—Cybersecurity and Penetration Testing—including complete outlines, module sequencing, and learning objectives.
- Recorded 24 video lectures for each course (48 total), delivering practical, hands-on instruction for a government skills initiative.
- Created quizzes, assignments, final projects, and exams to assess student progress and applied security competency.

Cyber Security Society – UMT

Cybersecurity Mentor

Feb 2025 – Jul 2025

Lahore, Pakistan

- Led weekly workshops on network security, threat modeling, and defensive security practices.
- Ran hands-on labs with Nmap, Wireshark, and Burp Suite to connect theory to real-world scenarios.
- Mentored students and grew an active campus security community focused on careers in information security.

Professional Freelancing Training Program

Cybersecurity Intern

Nov 2024 – Apr 2025

Lahore, Pakistan

- Tested web apps for SQL injection, file inclusion, and unrestricted upload vulnerabilities.
- Used Nmap, Burp Suite, and Wireshark for reconnaissance and traffic analysis; lab-tested EternalBlue, macro exploits, and emerging CVEs.
- Configured WAF and pfSense firewalls; performed digital forensics on hard drives and disk images.

RELEVANT PROJECTS

Nessus Vulnerability Assessment Lab

Tech Stack: Nessus, pfSense, Kali Linux, Windows 10, VirtualBox

Security Assessment

- Built an isolated assessment lab with pfSense, a Windows 10 VM exposing FTP, RPC, SSH, Telnet, RDP, and MySQL services, and Kali Linux; documented network topology and performed host/service reconnaissance.
- Configured Nessus and executed vulnerability scans; identified SSL/TLS issues (medium-strength ciphers, self-signed/untrusted certificates, deprecated TLS 1.0/1.1), SMB signing not required, and SSH Terrapin weakness (CVE-2023-48795).
- Assessed impact per finding, triaged false positives vs. confirmed vulnerabilities, and documented remediation recommendations with scan proof for validation.

ModSecurity WAF Lab

Tech Stack: DVWA, Apache, ModSecurity, mod_security2

Web Application Security

- Deployed DVWA and successfully exploited command injection to establish a vulnerable baseline before defensive controls were applied.
- Enabled Apache mod_security2, configured ModSecurity with rules cloned from GitHub, and integrated them in `apache2.conf`.
- Retested the same command injection payload after WAF deployment; ModSecurity blocked the attack and returned HTTP 403 Forbidden.

LOKI Scanner — Malware Detection Lab

Tech Stack: Apache, LOKI, YARA, Python, Kali Linux

Threat Detection

- Set up an Apache server with a file upload page, uploaded a web shell, and validated remote shell access to simulate a compromised host environment.

- Installed YARA, cloned LOKI from GitHub, configured the Python environment, and ran LOKI against the Apache working directory.
- Used LOKI to scan for malicious files and validate detection of the planted web shell in the target directory.

CTF Platform

Web Application

Tech Stack: Django, Docker, Bootstrap, Jinja2

- Built a Django web platform with an admin panel to add, edit, and remove CTF tasks—supporting both static-file and Docker-based challenges—with secure login and team creation.
- Implemented score calculation based on task marks and per-team task completion, with free and paid hint options for progressive challenge support.
- Delivered the participant frontend with Bootstrap and Jinja2; used to host CTF competitions at PFTP and lead Cyber Siege team events.

File Sync System

C++ Application

Tech Stack: C++, WinSock, AVL Tree, Multithreading

- Built separate C++ client and server applications on Windows using WinSock, implementing the server network stack from scratch for remote file management.
- Used multithreading so the main server process keeps accepting connections while each client request is handled on its own thread.
- Loaded millions of files into heap memory and indexed them with an in-memory AVL tree for fast search, list, create, delete, and modify operations.

POS System

Professional Work

Electron.js point-of-sale system for restaurants with order management, multi-printer support, and production deployment.

Voice Agent

AI Agent

Built production voice agents with Pipecat AI, Deepgram STT, OpenAI, and Cartesia TTS; implemented tool calling for transfers, voicemail detection, and API integrations.

Portfolio: talhazaheer.dev/#projects

ACHIEVEMENTS

- **Digital Pakistan Cybersecurity Hackathon** — Regional Round Participant (Red Teaming); Finalist (Blue Teaming)
- **PFTP / Cyber Siege** — Organized Capture The Flag competitions; led team to conduct campus CTF events
- **UMT** — Peer Tutor supporting cybersecurity and computer science coursework

EDUCATION

University of Management and Technology (UMT)

Bachelor of Science, Computer Science

Nov 2019 – Aug 2025

Lahore, Pakistan

Professional Freelancing Training Program (NIAIS)

Diploma, Cyber Security

Jun 2024 – May 2025

Lahore, Pakistan

NAVTTTC

Certification, Digital Forensic & Cyber Security

Apr 2025 – Jul 2025

Pakistan